CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

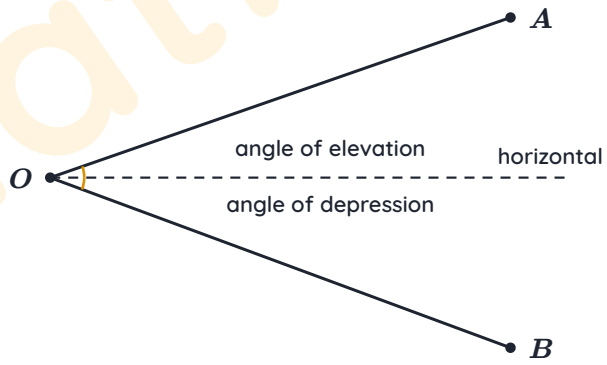
Chapter 9: Some Applications of Trigonometry

Topic-wise short notes with solved examples

Trigonometric ratios let us find heights and distances that cannot be measured directly, using an angle of elevation or depression and one known length.

1 Line of sight and the two angles

- The **line of sight** is the line drawn from the eye of the observer to the point of the object being viewed.
- The **angle of elevation** is the angle between the horizontal and the line of sight when the object is **above** the horizontal level. The observer looks up.
- The **angle of depression** is the angle between the horizontal and the line of sight when the object is **below** the horizontal level. The observer looks down.
- The angle of depression of a point B seen from A equals the angle of elevation of A seen from B, because they are alternate angles between parallel horizontal lines.



Both angles are measured from the horizontal through the eye O

REMEMBER

Both angles are measured from the **horizontal**, never from the vertical.

2 Choosing the ratio

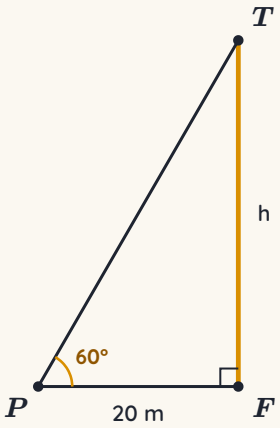
- Draw a figure. Mark the right angle, the given angle and the known length.
- Name the side you know and the side you want, relative to the given angle.

Sides involved	Ratio to use
Opposite and adjacent (height and ground distance)	$\tan$
Opposite and hypotenuse (height and slant length)	$\sin$
Adjacent and hypotenuse (ground distance and slant length)	$\cos$

- Values used most often: $\tan 30^\circ = \dfrac{1}{\sqrt{3}}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$. Take $\sqrt{3} = 1.732$ unless told otherwise.

3 One right-angled triangle

Example 3.1 The angle of elevation of the top of a tower from a point on the ground 20 m from its foot is 60°. Find the height of the tower.



Tower FT seen from the point P

SOLUTION

Let the height be h m. The height is opposite to the angle and the 20 m is adjacent.

$\tan 60^\circ = \dfrac{h}{20}$, so $h = 20\sqrt{3}$

Height $= 20\sqrt{3} \approx 34.64$ m

Example 3.2 A ladder 10 m long leans against a wall and makes an angle of 60° with the ground. Find the height it reaches on the wall and the distance of its foot from the wall.

SOLUTION

The ladder is the hypotenuse. Height: $\sin 60^\circ = \dfrac{h}{10}$, so $h = 10 \times \dfrac{\sqrt{3}}{2} = 5\sqrt{3}$ m

Distance: $\cos 60^\circ = \dfrac{d}{10}$, so $d = 10 \times \dfrac{1}{2} = 5$ m

Height $= 5\sqrt{3} \approx 8.66$ m; distance $= 5$ m

Example 3.3 From the top of a building 60 m high, the angle of depression of a car on the ground is 30°. Find the distance of the car from the foot of the building.

SOLUTION

The angle of elevation of the top of the building from the car is also 30° (alternate angles).

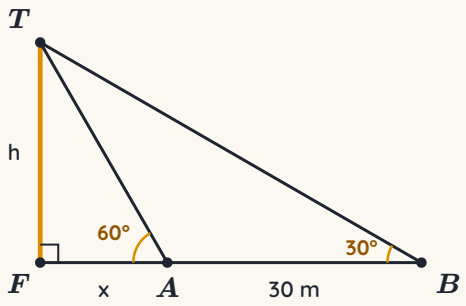
$\tan 30^\circ = \dfrac{60}{d}$, so $d = 60\sqrt{3}$

Distance $= 60\sqrt{3} \approx 103.92$ **m**

4 Two right-angled triangles

- Many problems have two triangles that share a side. Take the unknown shared side (or the unknown height) as x or h, write one equation from each triangle and solve.

Example 4.1 The shadow of a tower on level ground is 30 m longer when the sun's altitude is 30° than when it is 60°. Find the height of the tower.



Shadow FA when the sun is at 60° and FB when it is at 30°

SOLUTION

Let the height be h m and the shorter shadow be x m.

At 60°: $\tan 60^\circ = \dfrac{h}{x}$, so $h = x\sqrt{3}$

At 30°: $\tan 30^\circ = \dfrac{h}{x+30}$, so $x + 30 = h\sqrt{3} = 3x$

$x = 15$, so $h = 15\sqrt{3}$

Height $= 15\sqrt{3} \approx 25.98$ **m**

Example 4.2 From the top of a building 10 m high, the angle of elevation of the top of a tower is 45° and the angle of depression of its foot is 30°. Find the height of the tower.

SOLUTION

Let the horizontal distance between the building and the tower be d m.

Depression of the foot: $\tan 30^\circ = \dfrac{10}{d}$, so $d = 10\sqrt{3}$ m

Elevation of the top: the tower rises y m above the level of the roof, where $\tan 45^\circ = \dfrac{y}{d}$, so $y = 10\sqrt{3}$ m

Height of the tower $= 10 + 10\sqrt{3} = 10(1 + \sqrt{3})$

Height ≈ 27.32 m

Example 4.3 The angles of elevation of the top of a tower from two points on the same side of it, 5 m and 20 m from its base and in the same straight line with it, are complementary. Find the height of the tower.

SOLUTION

Let the height be h m and the angles be θ and $90^\circ - \theta$.

From the nearer point: $\tan\theta = \dfrac{h}{5}$

From the farther point: $\tan(90^\circ - \theta) = \cot\theta = \dfrac{h}{20}$

Multiply: $\tan\theta \times \cot\theta = \dfrac{h^2}{100}$, so $h^2 = 100$

Height $= 10$ m

Example 4.4 A boy 1.5 m tall stands 28.5 m from a building 30 m high. Find the angle of elevation of the top of the building from his eyes.

SOLUTION

Height of the top above eye level $= 30 - 1.5 = 28.5$ m

$\tan\theta = \dfrac{28.5}{28.5} = 1$

The angle of elevation is 45°.

5 Steps for every problem

- Read the problem and draw a neat figure. Label every point.
- Mark the angles of elevation and depression from the horizontal.
- Choose the unknowns and write a trigonometric equation from each right-angled triangle.
- Solve, give the answer with its unit, and round off only at the end.

6 Common mistakes

- Measuring the angle of depression from the vertical instead of the horizontal.
- Forgetting the height of the observer's eyes: subtract it before using the ratio, or add it back at the end.
- Using $\tan$ with the slant length. The slant length is the hypotenuse, so use $\sin$ or $\cos$.
- Rationalising too early and making arithmetic slips; keep $\sqrt{3}$ as it is until the last step.

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